

Lab 3, task #3 (11.05.2026)

Using the **pymoo (Python)** framework solve the following tasks.

- 1) Implement the solution for the following Multi-Objective Optimization (MOO) problem using the **NSGA II** algorithm (**max. 4 points**).

You are designing a cylindrical container for a new eco-friendly product. The optimization has two conflicting objectives:

- minimize the weight of the container (to reduce environmental impact and shipping costs),
- maximize the container volume (by minimization of the negative volume),

subject to a constraint as the crush strength of the container must exceed a certain threshold. The container is a closed cylinder with (decision variables):

- **radius** $r \in [2, 20]$ (in cm),
- **height** $h \in [20, 100]$ (in cm).

Let ρ — material density (assume 7.5 g/cm³), t — wall thickness in cm (assume 0.5 cm), the objective functions are:

1. **Weight** (in grams):

$$f_1 = \rho \cdot A \cdot t$$

$$A = 2\pi r h + 2\pi r^2$$

2. **Negative volume** (in cm³):

$$f_2 = -\pi r^2 h$$

Additionally, the container must resist a minimum crush force, thus the optimization constraint:

$$0.05 - \frac{r^3}{h} \leq 0$$

- 2) We also know that:

- the **production manager** supervising the construction of the container assumes the following weights assigned to optimization goals: container weight 55%, volume 45%,
- the **customer** (product recipient) assumes the following objective weights: container weight 65%, volume 35%.

Using the TOPSIS MCDM method (you can use the TopsisPy library), select from the non-dominated solutions obtained in 1. those that best meet the preferences of the – **max. 3 points**.

- a) production manager,
- b) customer.

- 3) Repeat the procedure described in 1. and 2. at least 10 times, then compare and contrast the final results from point 2. Plot the Pareto Front with all the points determined in 2. (divided by recipients) included on it – **max. 1 point**.